VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI-590 014

Major Project (BAI786) Report on

"AI-DRIVEN REAL TIME ANALYSIS OF AFI"

Submitted in partial fulfillment of the requirement for the award of degree of
BACHELOR OF ENGINEERING
IN
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted By

PAVAN G S	**- 4EG22AI029**
SANJAY H R	**- 4EG22AI037**
SUNIL T R	**- 4EG22AI045**
YASHAVANTHA N	**- 4EG22AI051**

Project Guide
Asst. Prof. Poornima S P M.Tech, Ph.d
Dept of CSE
GEC, Challakere

Project Coordinator
Asst. Prof. Khairunessa MCA, Ph.d
Dept of AIML,
GEC, Challakere

Head of the Department
Asst. Prof. Sahana V M.Tech, Ph.d
Dept of AIML
GEC, Challakere

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GOVERNMENT ENGINEERING COLLEGE CHALLAKERE

Ballary road, Challakere, Chitradurga-577522

(Affiliated To Visvesvaraya Technological University, Belagavi, Recognized By AICTE, New Delhi)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

CERTIFICATE

This is to Certify that the project, titled **"AI-DRIVEN REAL TIME ANLYSIS OF AFI"** carried out by **PAVAN G S(4EG22AI029), SANJAY H R(4EG22AI037), SUNIL T R(4EG22AI045), YASHAVANTHA N(4EG22AI051)** a Bonafide students of **Government Engineering College, Challakere** in partial fulfillment for the award of degree of **Bachelor of Engineering in Artificial Intelligence and Machine Learning of Visvesvaraya Technological University Belagavi** during the year 2025-26. It is certified that all correction/suggestions indicated for internal assessment have been incorporated in the Report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirement in respect of Project work prescribed for the said degree.

……...	……	…….
Signature of the Guide	Signature of the HOD	Signature of the Principal
Poornima S P M.Tech, Ph.d.	Sahana V M.Tech.,Ph.d	Dr. MM Benal Ph.D
Asst. Prof., Dept. of CSE	Asst. Prof., Dept. of AIML	Principal
GEC, Challakere	**GEC, Challakere**	**GEC, Challakere**

External Viva

Name of the Examiners	Signature with Date
1. …….	……
2. …….	……

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DECLARATION

We, **PAVAN G S(4EG22AI029), SANJAY H R(4EG22AI037), SUNIL T R(4EG22AI045), YASHAVANTHA N(4EG22AI051)** **the students of 7th semester Bachelor of Engineering in Artificial Intelligence and Machine Learning, GEC, Challakere, here by declare that the project report titled "AI-DRIVEN REAL TIME ANALYSIS OF AFI" has been prepared and presented by us and submitted in partial fulfilment of the course requirements for the award of degree in Bachelor of Engineering in Artificial Intelligence and Machine Learning, GEC, Challakere, during the year 2025- 2026. Further the content of the report has not been submitted previously for the award of any degree or diploma to any other university.**

Place: Challakere **Signature:**

Date:

ACKNOWLEDGEMENT

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Project Associates

PAVAN G S	**4EG22AI029**
SANJAY H R	**4EG22AI037**
SUNIL T R	**4EG22AI045**
YASHAVANTHA N	**4EG22AI051**

ABSTRACT

The challenges existing sleep monitoring system generates alarm sounds notification to the drivers when they are drowsy. They used computer vision method in that recognized face to detect drivers not eye. These are the current technology tensor flow, keras etc. While driving there with a lot of noise in the surrounding, the drivers can easily divert from sleep. Alert the drivers by message to take some rest for sometimes is useless while driving driver don't use the mobile then it is possible to see message.

To overcome this problem, the proposed system helps to prevent driver drowsiness. Then set a vibrate motor under a driver seat. It does not affect the driver in any manner. By using vibrate methodology, it can help the driver awake from the sleep these are the tools used in this project raspberry pi, pi camera, and vibration motor. Video can store the live recordings in external storage. Even though a network issue can prevent data from being stored if the car is in hilly area, external storage is provided here to record video. Rotary encoder will measure steering wheel. The car will automatically switch off if the steering radius is not consistent.

Keywords:

Amniotic Fluid Index (AFI) – Core medical parameter of the project, Obstetric Ultrasound / Prenatal – Domain context, Deep Learning / CNN – Model type used, Medical Image Analysis – Ultrasound processing focus, Real-Time AFI Prediction – Key contribution, Fetal Health Monitoring – Clinical relevance, Automated Ultrasound Analysis – Problem solved, Decision Support System – Doctor assistance aspect, TensorFlow, FastAPI – Technical stack (adds technical credibility), Web-Based Medical Application – Deployment nature

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